

Using Machine Learning to Predict NBA Player Statlines

Progress Report

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December, 2022

Abstract

With the increase of online statistics in sports, online betting has exploded with many different types of betting. In this study, we aim to create a machine learning model that can predict player stat betting in a consistent manner. The proposed solution will include three datasets and a number of different models that will be compared. Different models will be selected and developed. Once their performance is tested, hyper-parameter tuning will take place on the best-performing ones. Different betting techniques will be added to the models to determine which technique is the most efficient.

1 Introduction

The NBA is the biggest basketball league and one of the largest sports associations in the world, with 30 teams and around 450 players making up the entire league. Due to its popularity, there has been a rise in analytics and data collection with regard to teams and players. Basketball tends to be a *stat-heavy* sport as it is easy to keep track of every action and record it in a *stat sheet*.

1.1 Problem Definition

Sports betting means placing a wager on a sports event. Bookmakers will create *odds* for every outcome of an event depending on the likelihood of the outcome happening. A bet works by a bettor placing a bet on an outcome of a sports event that they think will happen. The *stake* is the amount of money that the bettor is comfortable putting on their bet. There are only two outcomes after the bet is placed the first being that the bettor *hit* (predicted correctly) their bet and they will get their stake back together with the profit (which is determined by the odds) and the second is when the bettor *misses* (predicts incorrectly) the bet and loses their money.

Suppose that the *LA Lakers* and the *Chicago Bulls*

will be playing against each other. The odds for the *LA Lakers* have 2.15 odds to win the game whilst the *Chicago Bulls* have 1.74 odds. The odds are usually denoted as follows:

LA Lakers 2.15 — 1.74 Chicago Bulls

The odds show the bettor that for every €1 bet, they will receive $1 \times \text{odds}$ if the bet hit (meaning they predicted the outcome correctly). In our example, if a bettor placed €10 on the *LA Lakers* winning and they do win, then the bettor will receive $10 \times 2.15 = 21.5$. This means that the bettor will get their *stake* back in addition to another €11.5 as pure profit.

The higher the odds of an outcome of a bet, the less likely to happen. In the previous example, the bookmakers thought that it was more likely for the *Chicago Bulls* to win (favourites) and that there was a higher chance of the *LA Lakers* losing the game (underdogs).

Put simply, in this project we will attempt to predict the outcome of such bets using one or more machine learning models. The data the model will be built on will be comprised of feature-rich datasets describing players and teams. The model will then be trained on data from previous NBA bets and their outcomes.

1.2 Motivation

In recent years, sports betting has exploded in popularity with the rise of online sports betting. Betting companies and bookmakers have all the power when it comes to setting the odds of an event and since they take a cut from every bet they will make money no matter the outcome. We hypothesise that the market has a vulnerability in a specific type of betting that can be exploited using machine learning. If we are successful, it will imply that the bookmakers' models for creating odds on this specific bet are not the most profitable and could be improved.

Additionally, we have not encountered any literature that indicates success over a lengthened period of time when it comes to this specific type of betting.

1.3 Aims and Objectives

The goal of this FYP is to identify and implement a machine learning algorithm that can effectively predict the outcome of bets over a long period of time. In order to achieve this goal, the following objectives have been identified:

1. Collecting, transforming and cleaning a dataset of previous and similar bets along with their outcome. This dataset will be used to train our models. No such dataset exists publicly, so we will employ techniques such as web scraping to gather this dataset.
2. Gathering player and team data on a game-by-game basis. The only games needed are the ones that have their respective bets.
3. Investigating a number of models using a sparse dataset and choosing a select few to further work on. Success will be measured on the accuracy that the model reaches. A benchmark model will be made to compare the accuracies.
4. Refining on the chosen models and attempting to develop them with the highest accuracy possible. This will be done with hyperparameter tuning.
5. Developing the models with different betting techniques and comparing profits. Some players play risky and some players play safe. We aim to find the most efficient way to bet.

1.4 Why is the problem Non-Trivial?

Creating a model that can rival the models used by bookmakers is a complex task due to the high-stakes nature of the betting business. We expect that the models used by bookmakers are close to the state of the art. Implementing this model will consist of complex data-gathering and data-cleaning techniques. The data will be manipulated with to experiment with the best possible dataset for the models.

1.4.1 Data Collection and Cleaning

There are three main datasets that will be needed to complete a model. The first is a dataset that will include previous similar bets. This data is not available to download, thus more advanced data-gathering techniques such as web scraping would have to be employed to get all the data needed. The second dataset includes every player's statistics from every game that they played in specific seasons. Finally, the third dataset is similar to the previous one, but instead of a player's statistics, it will hold the statistics of the team.

1.4.2 Model Selection

Another important task involves finding the best model that will support the requirements for our task. This will involve investigating various models and comparing them to one another. LSTMs are a form of neural network that fits this problem well and will be investigated and developed.

1.4.3 Testing Betting Techniques

There are many different ways that a bettor can choose to bet. Some bettors prefer to play it safe and bet big money on bets that are likely to hit. Others prefer to bet on multiple risky, unlikely bets, in the hope that one of them hits and makes up for the other bets as well. After the best model is identified, we will extend the model with different betting techniques to examine which technique is most efficient and compare techniques to one another.

2 Background and Literature Review

2.1 Player Betting

Bookmakers offer many different types of bets. In this FYP we will be focusing on player stat betting for NBA players. In every NBA game, there is a team known as the *scorers table* dedicated to keeping track of what happens in the game. For instance, if a player scores the ball then they will record it in the *stat sheet*. At the end of the game, every player will have totals of their stats throughout the game.

Bookmakers will release odds on how many stat totals each player will get every game. The format the bookmakers use to show the bet is using a number that we will call the *mid-point*. Player betting falls under a type of betting known as *Over/Under* betting, in which the bettor is provided with the mid-point and the only two options for betting are if the player will achieve over or under the mid-point.

Suppose that the *Lebron James* (an NBA player) will be playing a game. The bookmakers release the bet for his points that game. The mid-point is 26.5 points (the mid-point is always a decimal that ends in 5 so that the player can never finish with the same value as the mid-point) and the odds for him scoring under 26.5 points is 1.83 whilst scoring over is 1.9. This means that the bookmakers think that it is slightly more likely that *Lebron James* will score under the mid-point in this particular game. The bets are formatted in the following way:

Lebron James (Points) — Over 26.5 (1.9) / Under 26.5 (1.83)

Bookmakers will release these player bets for most of the players and most of the different types of stats in the league.

2.2 The Mathematics behind Sports Betting

2.2.1 Implied Win Probability

From the odds of a bet, one can calculate the win probability which is usually referred to as the *implied probability* [5]. To calculate the implied probability for a bet, we simply take the inverse of the decimal value for the odds [3]. The following formula demonstrates this:

$$\text{ImpliedProbability} = \frac{1}{\text{Odds}} \quad (1)$$

2.2.2 The Vigorish

Now that we can calculate the implied probability of any bet given its odds, we can calculate the implied probability of both sides of a bet. Since player betting only has two outcomes, i.e. either the player goes over or under the midpoint, we can safely assume that totalling the implied probability of both sides would add to 100%. In this example, we will use the aforementioned bet and equation.

Calculating Implied Probability for Lebron James (Points) — Over 26.5 (1.9)

$$\begin{aligned} \text{ImpliedProbability} &= \frac{1}{1.9} \\ \text{ImpliedProbability} &\approx 0.52631578947\% \end{aligned}$$

Calculating Implied Probability for Lebron James (Points) — Under 26.5 (1.83)

$$\begin{aligned} \text{ImpliedProbability} &= \frac{1}{1.83} \\ \text{ImpliedProbability} &\approx 0.54644808743\% \end{aligned}$$

If we total these two implied probabilities, we get *1.0727638769*. This value is not concurrent with the laws of probability, as $P(A) + P(A^c) = 1$, where A is the probability of an event happening and A^c is the complement of said event.

The reason why the implied probability is not accurate using the bookmakers' odds is due to the *Vigorish*. The vigorish is the percentage removed from a bettor's winnings by the bookmakers [2]. In this case, the vigorish of the bet calculated above is approximately *7.28%*, as

this is the extra percentage over *100%* when tallying the implied probabilities. The *overround* is a term coined to reference the summed implied probabilities greater than 1, which in this example would be *107.28%*.

2.2.3 Removing the Vigorish

In order to get the actual win probabilities based on the models of the bookmakers, we would have to remove the vigorish that is baked into their odds when they release them. Since probability theory claims that all possible events should equate to 1, we just need to scale the implied probabilities so that they sum up to 1. Dividing the implied probability by the overround will be equal to the actual probability.

$$\text{Actual Probability} = \frac{\text{Implied Probability}}{\text{Overround}} \quad (2)$$

The following are the calculations to gather the actual probabilities for the previous bet.

Actual Probability for Over ≈ 0.49061662198

Actual Probability for Under ≈ 0.50938337801

$$0.49061662198 + 0.50938337801 \approx 1$$

It is important to note that these are not the true probabilities of the events happening, as it is impossible to calculate the true probability. This is simply an educated guess based on the odds given to us by the bookmakers.

2.2.4 The Significance of 52.4%

Suppose you place a €10 wager on a 50/50 bet and win, you would expect to double your earnings. However, as previously mentioned, bookmakers will always take their cut. Normally, for an even-odds (50/50) game, the bookmaker will take 10% of the winnings [5]. Therefore, in order to make €10 on a coin flip, one would need to bet €11 (essentially a bet with 1.91 odds).

To find out what percentage of even-odd bets we would have to win on a consistent basis in order to be profitable, we straightforwardly plug the 1.91 odds into the formula for implied probability [5].

Implied Probability for bet with 1.91 Odds

$$\begin{aligned} \text{ImpliedProbability} &= \frac{1}{1.91} \\ \text{ImpliedProbability} &\approx 0.52356020942\% \end{aligned}$$

This indicates that for a bettor to break even on their even-odd bets, they would have to win 52.4% of the time to overcome the vigorish of the bookmakers.

2.2.5 The Balanced Book Hypothesis

The *Balanced Book Hypothesis* indicates that bookmakers aim to equally split the amount of money that is wagered on both sides of any even-odds bet [4]. It would be impossible for the bookmakers to lose money in this scenario as no matter which side wins, they will gather the profits from the bettors that lost and they take their vigorish from the winners as a cut.

In the case of player betting, bookmakers aim to achieve the balanced book hypothesis by adjusting the mid-point in their player bets. For instance, in the previously mentioned bet of *Lebron James*, if the mid-point was lower than 26.5 then chances are that more people would have bet on the over bet. In a situation where there is more money on one side, the bookmaker is vulnerable to losing money [4].

2.3 Literature Review

In Hubáček et al. [7], the goal was to predict a type of basketball betting known as *moneyline*. Moneyline bets involve predicting which team will win a game. If a team is much clearly better than their opponent, the odds will be more lopsided. For instance, the *Chicago Bulls* are playing the *LA Lakers*. From the current season, the *Chicago Bulls* have been playing better and have a better record for the season. Thus, their odds will be placed at 1.38 since they are the favourites and the *LA Lakers*' odds would be 3.15, implying that they are the underdogs.

The data they gathered consisted of various quantitative measures from both teams in their matches for that season. When they are predicting the outcome of a game, only the games that happened before the date of said game will apply. They used a *Logistic Regression* model as a baseline prediction model. They then developed two variants of a neural network. Firstly, a standard feed-forward network, while the second used a convolutional layer in tandem with three dense layers. The convolutional layer is specifically designed to deal with player-related data due to the high number of features.

The authors also proposed whether it is beneficial to not include the bookmakers' odds in the models themselves. If the odds are included as a feature then it is more likely that the predictions will be similar to the bookmakers'. Therefore, they did not include the odds in their models in order to *decorrelate* the models from the models of the bookmakers.

Testing was done over 9093 games that were played from 2006 — 2014. From their results, is it clear to see that it is to the detriment of the better to include the bookmakers' odds in the models. The convolutional neural network achieved higher accuracies and profits

than the rest of the models. Their results ended with the neural models being slightly less accurate than those of the bookmakers, whilst the logistic regression baseline was the least accurate one.

In Bucquet and Sarukkai's work [1], they set out to beat the bookmaker's in a different type of NBA betting known as *over/under*. This type of betting is more similar to player betting as the odds are always made to be as even as possible (such as in balanced book hypothesis) and involves predicting the total amount of points scored in a game. The bookmakers will provide the mid-point and bettors will bet on if they think both teams combined will score more or less than the mid-point.

For their features, they decided to look at each team's past three games and gather simple features such as points scored and points conceded, as well as other, more complicated features. To account for the opponent's strength, they added each opponent's season averages in certain selected stats. Finally, they included the number of days since the last game and the distance travelled to account for player fatigue. The data collected ranged from six seasons (2012 — 2018). Using the first four of these seasons as a testing set, 2016 — 2017 as a validation set and tested on the 2017 — 2018 season.

They investigated the following models: random forest, collaborative filtering, neural network and LSTM network. The model evaluation was done using mean squared error (MSE) between the over/under value predicted by each model and the actual point outcome of the game. In the end, the best-performing model was the neural network with an accuracy of 51.5% on average.

Papageorgiou [8] set out to solve the problem of predicting *fantasy points* for each player daily. Fantasy points is a single number that describes a player's performance in a single game.

The data gathered was from scraping the *nba.com* website and the writer proposed two main datasets, one for player statistics and one for team statistics. For the player dataset, they have gathered the individual statistics for each player in every game. Similarly, for teams, each team has a stat line for each game they played. Both datasets include each season between 2011 and 2021. The player dataset had 106 columns each representing a different stat, whilst the team dataset had 100 columns.

Each player had their own model made for them and after training, the models that suited the most players were random forest, voting regressor and the bayesian ridge. The evaluation was based on the actual fantasy points that a player scored as opposed to the predictions made by the models. The final unseen mean absolute percentage error (MAPE) was only 0.3% on the 2020-2021 dataset.

Donnelly [4] asked an important question: are the betting markets efficient? They are asking if the odds given to us by the bookmakers are optimal or not. The Efficient Market Hypothesis (EMH) was originally made for the financial markets and assumes that all publicly available information is reflected in the price of an investment. This was hard to prove in the financial world, but researchers have found that betting markets have adequate conditions to test the hypothesis since the problems are easier to determine [6].

The investigated type of betting is a *spread* bet in the National Football League (NFL). A spread bet involves a game between two teams that are not equally matched in skill. The underdog will get a handicap to even out the playing field. As an example, the *Philadelphia Eagles* and the *New York Giants* (two NFL teams) will be playing against each other. Since the Eagles are a better team, the Giants will get a +7 handicap. For betting purposes, this adds 7 points to the Giant’s final score. When a bettor bets on the Eagles, they think that the Eagles will win by more than 7 points. If a bettor places money on the Giants, they think that the Giants will either win, draw or lose by less than 7 points. In the case that the Giants lose by exactly 7 points, this results in a *push* and the stake that the bettor placed is returned.

The two main categories of information that were gathered for the paper were (1) point spreads and the actual outcome of each game and (2) the adjusted statistics for each game. A type of team evaluation known as *adjusted statistics* compares each play to a baseline that is representative of the league as a whole while accounting for the opponent’s difficulty. The seasons that were included in this paper were from the 2003-2004 season to the 2011-2012 season.

Instead of using machine learning to predict the outcome of a spread, they decided to use a weighted equation that takes both teams’ previous games played and how many points they score and conceded into account. Using the equation to determine the result of a spread, they managed to reach a 52.22% accuracy over three seasons, just under the 52.4% needed to beat the bookmakers.

3 Proposed Solution

3.1 Data Gathering

There will be three main datasets needed for this FYP. The first is for previous player bets alone with their outcome. This is arguably the hardest data to collect, as this type of betting is relatively new and there is not much attention put on it. With this being said, we have found a website that publishes different player bets from different bookmakers. Whilst the bets are not simply

available to download, it is possible to make use of web scraping.

The second and third datasets are similar in nature and can be gathered together. An application programming interface (API) is available online that provides different information on previous NBA matches. By combining certain endpoints in the API, we can gather most of the information needed in order to train the model.

3.2 Choosing and Training Models

After the data has been gathered, we will investigate different models that fit the requirements and could potentially work. For example, LSTMs worked well in the previous, similar literature and we intend on developing a variation of one. After finding a shortlist of these models, we will develop them and tune their hyper-parameters to get the best accuracy possible when testing whilst also avoiding overfitting.

The aim is to create a confidence score for each possible bet in the hopes that each model is able to find the biggest vulnerabilities when it comes to the odds of the bookmaker.

3.3 Betting Techniques

Bettors have different strategies when playing that differ between each player. Some players prefer to play it safe and build up their profits slowly, whilst others prefer the riskier route and play small numbers in the hopes to return big.

We will test out different strategies in the hopes of finding one that is the most efficient and that can potentially return the most profits. Another form of betting technique could come in the form of having a confidence score on every potential bet of that day and the model could for example, pick the top five most confident bets and place an equal amount or varying amounts depending on how confident the model is in a specific bet.

3.4 Evaluation Plan

The evaluation begins with the data gathering. It is hard to pinpoint a certain number of records of bets that are needed to perfectly train each model. So far we have gathered just over eighteen thousand previous player bets from the 2021 — 2022 season. We will train each model using this number of records. We will attempt to configure the models to work more efficiently with a sparse dataset.

The second stage of evaluation comes in the form of grading the different models that will be used. Model evaluation is more on the simple side as they can simply be ranked according to the accuracy that they achieve

when predicting the outcome of the bets. If the confidence score is used, we can measure success using a mean square error function or a similar technique.

The final step to evaluate is the different betting techniques. Every model that was chosen to expand upon will be equipped with all the betting strategies that will be used. The different techniques can then be ranked according to the profits that they yield.

4 Conclusions

The final goal is to understand the betting market better and show that there are vulnerabilities in the market that the bookmakers are not noticing and that can be exploited. The expected challenges for this project are to deciding exactly which models to use and attempt to pick the models that will have the best results after hyperparameter tuning.

We are expecting that finding the right hyperparameters will also be difficult. We will use a grid search to test different configurations of hyperparameters such as learning rate.

Training time is also a concern as we do not have an estimate on how long each model will take to train with all the data that is going to be used. We will need to create an efficient algorithm in order to save on any training time that we can.

There is also the issue of potentially not having enough data, we are limited to the records that we have for now and we are not sure that all the models will be trained well with a sparse dataset.

Finally, we have encountered no literature that is attempting to solve the same problem as this, so we have no direct blueprint to follow.

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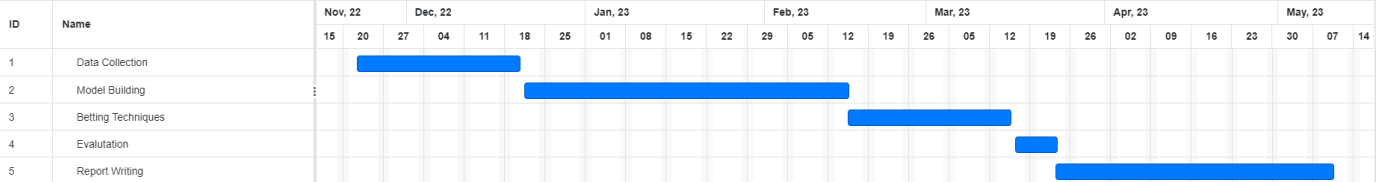


Figure 1: Gantt chart of project